SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	PRIYADHARSHINI K S	**Reg. No.:**	192511248
Date:	31/07/2026	**Duration:**	60 minutes

Code of Conduct

I _PRIYADHARSHINI - 192511248______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ____PRIYADHARSHINI K S_____________________

Vehicle Service Scheduling and Predictive Maintenance Management System

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. Identify the Project Vision and Stakeholders

Project Vision:

The vision of the Vehicle Service Scheduling and Predictive Maintenance Management System is to provide an efficient and user-friendly platform for managing vehicle servicing activities digitally. The system aims to automate service scheduling, maintain digital maintenance records, and send timely service reminders to customers. It reduces manual paperwork, minimizes human errors, and improves communication between customers and service centres. The system also helps service centres manage appointments efficiently while improving customer satisfaction and overall service quality.

Objectives

- Automate vehicle service scheduling.
- Maintain digital service records.
- Provide online appointment booking.
- Send automatic maintenance reminders.
- Reduce paperwork and manual errors.

- Improve service centre efficiency.
- Enable easy tracking of vehicle service history.
- Ensure secure customer and vehicle data management.
- Improve customer satisfaction.
- Support future system enhancements

Stakeholders:

Vehicle Owner

- Registers vehicle details.
- Books service appointments.
- Receives service reminders.
- Views maintenance history.
- Tracks service status.

Service Centre Manager

- Manages appointments.
- Assigns mechanics.
- Monitors service progress.
- Generates reports.
- Oversees daily operations.

Mechanic

- Inspects vehicles.
- Performs servicing and repairs.
- Updates maintenance records.
- Views previous service history.
- Marks service completion.

System Administrator

- Manages users.
- Maintains database security.
- Performs system backup.
- Generates reports.
- Monitors system performance.

2. Create Epics and User Stories

Epic 1: User Registration & Authentication

- As a vehicle owner, I want to register an account, so that I can access the vehicle service management system.
- As a vehicle owner, I want to log in securely, so that I can book and manage my service appointments.

Epic 2: Vehicle Management

- As a vehicle owner, I want to register my vehicle details, so that my vehicle information is stored securely.
- As a vehicle owner, I want to view my vehicle maintenance history, so that I can track previous services.

Epic 3: Service Booking

- As a vehicle owner, I want to book a service appointment online, so that I can avoid waiting at the service centre.
- As a service centre manager, I want to approve and manage service bookings, so that appointments are scheduled efficiently.

Epic 4: Predictive Maintenance & Notifications

- As a vehicle owner, I want to receive automatic service reminders, so that I never miss a scheduled maintenance.
- As a vehicle owner, I want to receive maintenance alerts, so that I can prevent unexpected vehicle breakdowns.

Epic 5: Service Management

- As a mechanic, I want to update the service status, so that customers can track their vehicle servicing progress.
- As a mechanic, I want to update maintenance records, so that the vehicle service history remains accurate.

Epic 6: Administration & Reports

- As a system administrator, I want to manage users and system data, so that the application remains secure and organized.
- As a service centre manager, I want to generate service reports, so that I can monitor service centre performance and make informed decisions.

3. Define Acceptance Criteria

User Registration

Given the user is on the registration page, **When** the user enters valid personal details and clicks Register, **Then** a new account should be created successfully and the user should receive a confirmation message.

User Login

Given the user has a registered account, **When** the user enters a valid username and password, **Then** the system should authenticate the user and display the dashboard.

Vehicle Registration

Given the user is logged into the system, **When** the user enters valid vehicle details and submits the form, **Then** the vehicle information should be stored successfully in the database.

View Maintenance History

Given the vehicle has previous service records, **When** the user selects the maintenance history option, **Then** the system should display all previous service details correctly.

Book Service Appointment

Given the user is logged in, **When** the user selects a service date and confirms the booking, **Then** the appointment should be scheduled successfully and a confirmation message should be displayed.

Manage Service Booking

Given service booking requests are available, **When** the service centre manager approves or updates the booking, **Then** the booking status should be updated and visible to the customer.

Service Reminder Notification

Given a service date is approaching, **When** the scheduled reminder time is reached, **Then** the system should automatically send a reminder notification to the customer.

Predictive Maintenance Alert

Given the vehicle maintenance data is available, **When** the system detects that servicing is due, **Then** it should notify the customer with a maintenance alert.

Update Service Status

Given the mechanic has completed the service, **When** the mechanic updates the service details, **Then** the system should save the updated service status and notify the customer.

Generate Reports

Given service records are available, **When** the manager requests a report, **Then** the system should generate an accurate service report for viewing or download.

4. Prioritize the Product Backlog

User Registration – Must Have:

Essential for creating user accounts and accessing the system.

User Login – Must have:

Required for secure authentication and system access.

Vehicle Registration – Must have:

Stores vehicle details required for service management.

Book Service Appointment – Must Have:

Allows customers to schedule vehicle servicing online.

Manage Service Bookings – Must Have:

Enables managers to approve and organize appointments efficiently.

View Maintenance History – should have:

Helps customers track previous vehicle services.

Service Reminder Notification – should have:

Sends reminders to customers before service due dates.

Update Service Status – should have:

Allows mechanics to update completed service details.

Predictive Maintenance Alert – could have:

Provides maintenance alerts based on vehicle service history.

Generate Reports – could have:

Generates reports for monitoring service centre performance.

5. Plan the Sprint Backlog

User Story	Development Tasks
User Registration	Design registration page, create registration form, validate user inputs, store user data in database, test registration functionality.
User Login	Design login page, implement authentication, verify user credentials, create session management, test login process.
Vehicle Registration	Create vehicle registration form, validate vehicle details, store vehicle information in database, test vehicle registration module.
Book Service Appointment	Design booking interface, implement appointment scheduling, save booking details, display booking confirmation, test booking functionality.
Manage Service Booking	Create manager dashboard, view booking requests, approve/reject appointments, update booking status, test booking management features.

6. Estimate Effort Using Story Points

User Story	Story Points	Justification
User Registration	**3**	Simple user registration form with database storage and validation.
User Login	**3**	Requires user authentication and secure session management.
Vehicle Registration	**5**	Involves form validation, database integration, and linking vehicles to users.
Book Service Appointment	**8**	Includes appointment scheduling, date/time selection, and booking confirmation.
Manage Service Bookings	**8**	Requires manager dashboard, booking approval, status updates, and notifications.

7. Present the Sprint Goal and Expected Deliverables

Sprint Goal:

The goal of Sprint 1 is to develop the core functionalities of the Vehicle Service Scheduling and Predictive Maintenance Management System. This sprint focuses on enabling users to register, log in securely, register their vehicles, and book service appointments online. It also allows the service centre manager to manage customer bookings efficiently. By the end of the sprint, the team aims to deliver a stable and functional system with the essential features required for basic vehicle service management. This provides a strong foundation for future development and enhancements.

Expected Sprint Deliverables

- User Registration Module completed.
- Secure User Login Module implemented.
- Vehicle Registration Module developed.
- Online Service Booking feature completed.
- Service Booking Management Dashboard for managers.
- Database created for users, vehicles, and bookings.
- Basic user interface with navigation completed.
- Core functionalities tested and validated.
- Working prototype ready for stakeholder review.
- Sprint review and feedback collected for the next sprint

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable,	Acceptance criteria are mostly clear and	Acceptance criteria are partially defined	Acceptance criteria are missing or not

	testable, and aligned with all user stories.	testable with minor gaps.	and lack clarity or completeness.	aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	**Mark**
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25